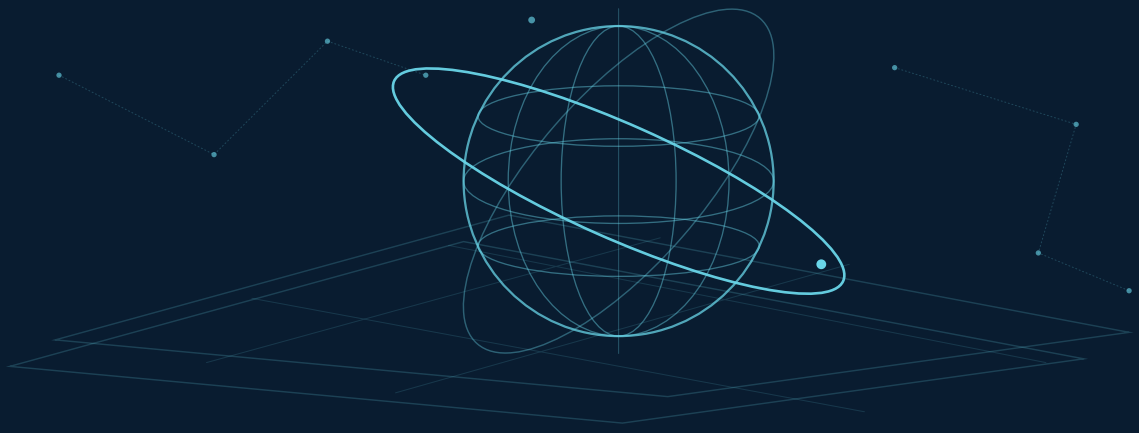


SPECIFICATION

Task board — create, complete, and list

Synthetic L1 reference for Atlas Specifications

DOC-TASK-BOARD

Revision 1 / draft / Alpha-v0.1.0

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Authors

- Douglas de Oliveira Alkimim

Specification author

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Context, scope, and evidence

This synthetic board demonstrates the Atlas L1 contract for creating, completing, and listing Tasks. It does not implement a business application or assert independent stakeholder validation.

Origin: SRC-TASK-BRIEF is the versioned teaching baseline. The three goals, rules, authorization policy and quality obligations are didactic design choices. No Needs-to-L1 derivation, interview, or named human approval is claimed.

Out of scope: editing titles, deleting Tasks, multiple boards, anonymous access, storage technology, authentication mechanisms, transport protocols, screens, and quantitative response-time targets.

ACTOR ACT-BOARD-MEMBER

Name

Board member

Kind: role

Description

An external person acting as a member of the selected board; concrete caller identity and permissions are resolved separately.

TYPE TYP-TASK-STATUS

Name

Task status

Definition

Lifecycle state of a Task.

Constraints

Exactly open or completed.

Examples

- open
- completed

Entities and canonical rules

Entities own meaning and validity; they have no collaborators.

ENTITY ENT-TASK

Name

Task

Meaning

A unit of work retained in one board. Its identity and title survive completion. The entity owns data validity and state-transition rules; it has no tool or use-case dependency.

Identity

Description

Tasks with exactly equal task identifiers in the same board are the same task.

Fields: FLD-TASK-ID

Fields

FLD-TASK-ID

Name

Task identifier

Type: text

Cardinality: 1

Required: true

Description

Identity within this board.

Constraints

Non-empty, non-whitespace text; equality is exact and case-sensitive.

Examples

Valid

- TASK001

Invalid

-

FLD-TASK-TITLE

Name

Title

Type: text

Cardinality: 1

Required: true

Description

The work to be remembered.

Constraints

At least one non-whitespace character; retain the original text.

Examples

Valid

- Review Atlas

Invalid

-

FLD-TASK-STATUS

Name

Status

Type: task-status

Cardinality: 1

Required: true

Description

Lifecycle state.

Constraints

One of open or completed.

Examples

Valid

- open

Invalid

- deleted

FLD-TASK-REVISION

Name

Revision

Type: integer

Cardinality: 1

Required: true

Description

Concurrency token for a retained Task.

Constraints

A non-negative integer, increased by one for each successful completion.

Examples

Valid

- 0

Invalid

- -1

Relationships: none

States

ST-TASK-OPEN

Name

Open

Description

Work is still pending.

ST-TASK-COMPLETED

Name

Completed

Description

Work has been completed; the task remains retained.

Transitions

TRN-TASK-COMPLETE

Name

Complete

From: ST-TASK-OPEN

To: ST-TASK-COMPLETED

Description

Preserve identity and title; change status to completed. The store increments revision after the conditional write succeeds.

Validations

VLD-TASK-ID

Name

Task identifier

Scope: field

Statement

A task identifier is non-empty text, preserved exactly.

Inputs

Name

Candidate

Source: input

Meaning

A task identifier is non-empty text, preserved exactly.

Conditions

C-TASK-ID-1

Condition

The candidate is text containing at least one non-whitespace character.

Meaning

The candidate is text containing at least one non-whitespace character.

Results

Outcome: pass

Description

All stated conditions hold. No normalization or persistence occurs.

Outcome: fail

Description

At least one stated condition fails. Preserve prior state.

Flow: GRAPH-VLD-TASK-ID

VLD-TASK-NEW

Name

New task validity

Scope: entity

Statement

A new Task has a valid identifier, a meaningful title, open status, and revision zero.

Inputs

Name

Candidate

Source: input

Meaning

A new Task has a valid identifier, a meaningful title, open status, and revision zero.

Conditions

C-TASK-NEW-1

Condition

The task identifier satisfies VLD-TASK-ID.

Meaning

The task identifier satisfies VLD-TASK-ID.

C-TASK-NEW-2

Condition

The title is text with at least one non-whitespace character.

Meaning

The title is text with at least one non-whitespace character.

C-TASK-NEW-3

Condition

The status is open.

Meaning

The status is open.

C-TASK-NEW-4

Condition

The revision is the integer zero.

Meaning

The revision is the integer zero.

Results

Outcome: pass

Description

All stated conditions hold. No normalization or persistence occurs.

Outcome: fail

Description

At least one stated condition fails. Preserve prior state.

Flow: GRAPH-VLD-TASK-NEW

VLD-TASK-COMPLETE

Name

Completion transition

Scope: transition

Statement

A ready Task may transition from open to completed; a completed Task cannot be completed again.

Inputs

Name

Candidate

Source: state

Meaning

A ready Task may transition from open to completed; a completed Task cannot be completed again.

Conditions

C-TASK-COMPLETE-1

Condition

The current state is ST-TASK-OPEN.

Meaning

The current state is ST-TASK-OPEN.

Results

Outcome: pass

Description

All stated conditions hold. No normalization or persistence occurs.

Outcome: fail

Description

At least one stated condition fails. Preserve prior state.

Flow: GRAPH-VLD-TASK-COMPLETE

Examples

EX-TASK-OPEN

Kind: valid

Description

A valid new Task.

Data

```
{
"task_id": "TASK001",
"title": "Review Atlas",
"status": "open",
"revision": 0
}
```

EX-TASK-COMPLETED

Kind: valid

Description

A retained Task after one completion.

Data

```
{
"task_id": "TASK001",
"title": "Review Atlas",
"status": "completed",
"revision": 1
}
```

EX-TASK-INVALID

Kind: invalid

Description

An empty title violates entity validity.

Data

```
{
"task_id": "TASK001",
"title": "",
"status": "open",
"revision": 0
}
```

Assumptions: ASM-TASK-SINGLE-BOARD

Open issues: none

Abstract tool contracts

Tools consume ready entities and define atomicity, failure distinctions and persistence uncertainty without selecting implementations.

TOOL TOL-TASK-STORE

Name

Task store

Purpose

An abstract retention and retrieval capability. This specification chooses no database, protocol, framework, SQL syntax, route, or concrete adapter.

Operations

OP-TASK-CREATE

Name

Retain a new Task

Purpose

Atomically insert one ready open Task if its identifier is absent.

Inputs

Name

Task

Type: ENT-TASK

Meaning

Ready Task after VLD-TASK-NEW; revision zero.

Required: true

Name

Board

Type: resolved-board

Meaning

The board bound to trusted caller context.

Required: true

Conditions

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

All operations act on the board resolved in the trusted caller context. A caller supplies no untrusted substitute for that board binding.

Results

Name

Created Task

Type: ENT-TASK

Meaning

The retained Task, with its exact title and identifier.

Guarantees

A successful result confirms retention at revision zero.

Effects

One Task is added on success.

Failures

Code: DUPLICATE_ID

Meaning

The identifier is already retained in this board.

Consumer distinction

This outcome is not applied: the store guarantees that this operation made no change.

Code: STORE_UNAVAILABLE

Meaning

The capability could not confirm the requested result.

Consumer distinction

For a write, return write_outcome=not_applied only with evidence no write occurred; otherwise return write_outcome=unknown. Unknown is neither success nor proof of no change, and must not be automatically retried. A read failure returns no usable snapshot and changes no task.

Guarantees

Uniqueness check and insertion are atomic. A duplicate does not overwrite an existing Task.

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

OP-TASK-LOAD

Name

Read one Task

Purpose

Return a ready Task and its observed revision by exact identifier.

Inputs

Name

Task identifier

Type: text

Meaning

Identifier already checked against VLD-TASK-ID.

Required: true

Name

Board

Type: resolved-board

Meaning

The authorized board.

Required: true

Conditions

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

All operations act on the board resolved in the trusted caller context. A caller supplies no untrusted substitute for that board binding.

Results

Name

Task

Type: ENT-TASK

Meaning

A ready Task hydrated from one retained record.

Guarantees

Its revision describes the same observed record as its fields.

Effects

No task is changed.

Failures

Code: NOT_FOUND

Meaning

No task with this identifier exists in this board at read time.

Consumer distinction

Distinct from unavailable or malformed storage.

Code: STORE_UNAVAILABLE

Meaning

The capability could not confirm the requested result.

Consumer distinction

For a write, return write_outcome=not_applied only with evidence no write occurred; otherwise return write_outcome=unknown. Unknown is neither success nor proof of no change, and must not be automatically retried. A read failure returns no usable snapshot and changes no task.

Code: STORE_INVALID_STATE

Meaning

A stored record could not be hydrated through the canonical Task validity rules.

Consumer distinction

Do not return a malformed ready Task, an empty success, or a partial list. Report the integrity failure; no task is changed by the read.

Guarantees

No write occurs. Not-found is a confirmed absence, not a masked read error.

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

OP-TASK-COMPLETE

Name

Retain a completion

Purpose

Atomically replace the observed Task with a ready completed Task only if its revision still matches.

Inputs

Name

Task

Type: ENT-TASK

Meaning

Ready Task after the entity's open-to-completed transition; identity/title unchanged.

Required: true

Name

Expected revision

Type: integer

Meaning

Revision returned by OP-TASK-LOAD.

Required: true

Name

Board

Type: resolved-board

Meaning

The authorized board.

Required: true

Conditions

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

All operations act on the board resolved in the trusted caller context. A caller supplies no untrusted substitute for that board binding.

Results

Name

Completed Task

Type: ENT-TASK

Meaning

The retained completed Task.

Guarantees

Revision is expected_revision+1; identity and title are unchanged.

Effects

Status becomes completed and revision increases once on success.

Failures

Code: NOT_FOUND

Meaning

The observed Task is no longer present at conditional-write time.

Consumer distinction

This outcome is not applied: the store guarantees that this operation made no change.

Code: VERSION_CONFLICT

Meaning

The retained revision differs from the expected revision.

Consumer distinction

This outcome is not applied: the store guarantees that this operation made no change.

Code: STORE_UNAVAILABLE

Meaning

The capability could not confirm the requested result.

Consumer distinction

For a write, return write_outcome=not_applied only with evidence no write occurred; otherwise return write_outcome=unknown. Unknown is neither success nor proof of no change, and must not be automatically retried. A read failure returns no usable snapshot and changes no task.

Guarantees

Existence/revision comparison and replacement are atomic; no lost update is reported as success.

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do

not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

OP-TASK-LIST

Name

Read the board

Purpose

Return one consistent snapshot of all Tasks in the authorized board.

Inputs

Name

Board

Type: resolved-board

Meaning

The authorized board.

Required: true

Conditions

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

All operations act on the board resolved in the trusted caller context. A caller supplies no untrusted substitute for that board binding.

Results

Name

Tasks

Type: list[ENT-TASK]

Meaning

Ready Tasks ordered by exact identifier using Unicode code-point order.

Guarantees

Each matching Task appears exactly once; an empty successful list means the board was empty at the snapshot instant.

Effects

No task is changed.

Failures

Code: STORE_UNAVAILABLE

Meaning

The capability could not confirm the requested result.

Consumer distinction

For a write, return write_outcome=not_applied only with evidence no write occurred; otherwise return write_outcome=unknown. Unknown is neither success nor proof of no change, and must not be automatically retried. A read failure returns no usable snapshot and changes no task.

Code: STORE_INVALID_STATE

Meaning

A stored record could not be hydrated through the canonical Task validity rules.

Consumer distinction

Do not return a malformed ready Task, an empty success, or a partial list. Report the integrity failure; no task is changed by the read.

Guarantees

No partial list is returned as a complete snapshot. Concurrent later changes need not appear in this snapshot.

A Task input is already valid according to ENT-TASK. Tools consume ready entities and do not repeat entity-invariant validation. Loading raw records uses the canonical entity construction/hydration boundary once; this is not duplicate validation of a ready Task.

Use cases and acceptance

Each case declares actor context, functional behavior, flow links, quality obligations, and acceptance criteria.

USE CASE `UC-TASK-CREATE`

Name

Create task

Objective

Remember one new unit of work in the board.

Actors

Actor: ACT-BOARD-MEMBER

Permission

task.create

Description

Initiating actor: board member. Require the caller to be resolved to a principal with task.create for this board; a role name supplied as business input is insufficient.

Trigger

The board member (ACT-BOARD-MEMBER) requests: create task.

Context

Required caller context: resolved principal identifier, selected board identifier, authenticated status, granted case permissions, issuer, and validity status. The context originates from a trusted identity-and-permission authority configured for the application; claims in the task request are untrusted and cannot supply authorization.

The trust assumption is ASM-TASK-IDENTITY. Authentication is required; its mechanism and token/channel encoding belong to later design. Authorization is reevaluated for this invocation, and composed invocations would preserve the same principal and board context.

The actor is a role. The caller is the concrete principal for this request. Neither is an accountable human author for a Needs-to-L1 derivation.

Inputs

Name

Task identifier

Type: text

Meaning

Requested stable task identifier.

Required: true

Name

Title

Type: text

Meaning

The work to remember.

Required: true

Entry conditions

Requests may arrive with invalid or missing authorization context; the case's denial path remains defined. No downstream effect occurs before authorization succeeds.

Entities

Entity: ENT-TASK

Validations: VLD-TASK-NEW

Purpose

Use the canonical Task meaning and rules. Tools receive ready entities; no repeated validation of ready entity invariants is required.

Tools

Tool: TOL-TASK-STORE

Operation: OP-TASK-CREATE

Purpose

Use the declared capability and handle every stated observable outcome.

Calls: none

Outcomes

`OUT-TASK-CREATE-DENIED`

Kind: refusal

Description

The caller context is absent, unresolved, expired, or lacks this case's permission for this board.

Effects

No tool call, no write, and no disclosure of task existence or content.

`OUT-TASK-CREATE-CREATED`

Kind: success

Description

The retained Task is returned.

Effects

Exactly one open Task at revision zero is retained.

OUT-TASK-CREATE-INVALID

Kind: refusal

Description

The new-task validity rule failed.

Effects

No store invocation or state change.

OUT-TASK-CREATE-DUPLICATE

Kind: refusal

Description

The identifier was already retained.

Effects

No change to that Task.

OUT-TASK-CREATE-UNAVAILABLE

Kind: failure

Description

Storage did not confirm creation; report write_outcome.

Effects

not_applied guarantees no write; unknown leaves final retention state unresolved.

Functional requirements

RF-TASK-CREATE-AUTH

Statement

Resolve the initiating board member and require task.create for the selected board in trusted caller context. Missing, unresolved, expired, or insufficient context is denied before any entity application or tool call. No anonymous execution is permitted. Origin: SRC-TASK-BRIEF.

Scope: case

Applies to: UC-TASK-CREATE

Flow references: N-TASK-CREATE-AUTH, N-TASK-CREATE-DENIED, E-TASK-CREATE-DENIED, E-TASK-CREATE-ALLOWED

RF-TASK-CREATE-DATA

Statement

After authorization, construct a new Task from task_id and title with status open and revision zero, applying VLD-TASK-NEW. An invalid candidate is refused before storage; preserve input text exactly.

Scope: case

Applies to: UC-TASK-CREATE

Flow references: N-TASK-CREATE-VALIDATE, N-TASK-CREATE-VALID, N-TASK-CREATE-INVALID, E-TASK-CREATE-2

RF-TASK-CREATE-RETAIN

Statement

Pass the ready Task and authorized board to OP-TASK-CREATE once. Only a confirmed result returns the retained identifier, title, open status, and revision zero.

Scope: operation

Applies to: UC-TASK-CREATE, OP-TASK-CREATE

Flow references: N-TASK-CREATE-STORE, N-TASK-CREATE-CREATED, E-TASK-CREATE-3, E-TASK-CREATE-6

RF-TASK-CREATE-DUPLICATE

Statement

Map DUPLICATE_ID to duplicate-identifier refusal. Preserve the existing Task and do not choose a replacement identifier silently.

Scope: case

Applies to: UC-TASK-CREATE

Flow references: N-TASK-CREATE-DUPLICATE, E-TASK-CREATE-4

RF-TASK-CREATE-FAILURE

Statement

On STORE_UNAVAILABLE, report write_outcome supplied by the tool. Distinguish not_applied from unknown; never infer no change or success from an absent confirmation and do not retry automatically.

Scope: operation

Applies to: UC-TASK-CREATE, OP-TASK-CREATE

Flow references: N-TASK-CREATE-UNAVAILABLE, E-TASK-CREATE-5

Nonfunctional requirements

RNF-TASK-CREATE-RETENTION

Scope: system

Applies to: UC-TASK-CREATE, OP-TASK-CREATE, UC-TASK-LIST

Statement

A confirmed creation remains readable after a normal store restart. Origin: SRC-TASK-BRIEF (synthetic teaching brief).

Verifiable criterion

After a confirmed create and normal restart, a successful list returns the Task with unchanged fields exactly once.

Verification

Exercise create, restart the chosen persistence implementation normally, then list. This repository supplies the contract, not that implementation.

References: SRC-TASK-BRIEF

Applicability

Applies under the explicitly stated case and tool conditions. This is an obligation for a future implementation, not a measured claim about this repository.

Flows

Main: GRAPH-TASK-CREATE

Subflows: none

Flow text

Resolve trusted context first. Deny missing, unresolved, expired, or insufficient permission before validation and storage (RF-TASK-CREATE-AUTH).

Apply VLD-TASK-NEW to input data once. Refuse invalid input without tool access (RF-TASK-CREATE-DATA).

Pass the ready Task to OP-TASK-CREATE. Confirm only on its success; map duplicate and unavailable outcomes explicitly (RF-TASK-CREATE-RETAIN, RF-TASK-CREATE-DUPLICATE, RF-TASK-CREATE-FAILURE).

Acceptance criteria

AC-TASK-CREATE-AUTH

Statement

Try absent context, an unresolved identity, an expired context, and a member lacking this permission. Every attempt is denied before tool invocation, with no task state or existence disclosed.

Verifies: RF-TASK-CREATE-AUTH

AC-TASK-CREATE-SUCCESS

Statement

With trusted create permission and unused TASK001 / Review Atlas, receive the exact open Task at revision zero; a later successful list includes it once.

Verifies: RF-TASK-CREATE-DATA, RF-TASK-CREATE-RETAIN, RNF-TASK-CREATE-RETENTION

AC-TASK-CREATE-INVALID

Statement

An empty title or whitespace-only identifier is refused before OP-TASK-CREATE is invoked.

Verifies: RF-TASK-CREATE-DATA

AC-TASK-CREATE-DUPLICATE

Statement

Creating TASK001 twice retains the first Task unchanged and returns duplicate-identifier refusal on the second attempt.

Verifies: RF-TASK-CREATE-DUPLICATE

AC-TASK-CREATE-UNKNOWN

Statement

Inject a write with lost confirmation. The case reports unknown; it does not claim no write, report success, or initiate a retry.

Verifies: RF-TASK-CREATE-FAILURE

Traceability

TRC-TASK-CREATE-ORIGIN

Source: SRC-TASK-BRIEF

Targets: RF-TASK-CREATE-AUTH, RF-TASK-CREATE-DATA, RF-TASK-CREATE-RETAIN, RF-TASK-CREATE-DUPLICATE, RF-TASK-CREATE-FAILURE, RNF-TASK-CREATE-RETENTION

Note

Synthetic requirements selected for this teaching example; not evidence of stakeholder approval.

TRC-TASK-CREATE-AC-1

Source: AC-TASK-CREATE-AUTH

Targets: RF-TASK-CREATE-AUTH

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-CREATE-AC-2

Source: AC-TASK-CREATE-SUCCESS

Targets: RF-TASK-CREATE-DATA, RF-TASK-CREATE-RETAIN, RNF-TASK-CREATE-RETENTION

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-CREATE-AC-3

Source: AC-TASK-CREATE-INVALID

Targets: RF-TASK-CREATE-DATA

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-CREATE-AC-4

Source: AC-TASK-CREATE-DUPLICATE

Targets: RF-TASK-CREATE-DUPLICATE

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-CREATE-AC-5

Source: AC-TASK-CREATE-UNKNOWN

Targets: RF-TASK-CREATE-FAILURE

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

Assumptions: ASM-TASK-SINGLE-BOARD, ASM-TASK-IDENTITY

Sources: SRC-TASK-BRIEF, SRC-TASK-OAL

Open issues: none

USE CASE UC-TASK-COMPLETE

Name

Complete task

Objective

Mark an existing open Task completed without losing a competing change.

Actors

Actor: ACT-BOARD-MEMBER

Permission

task.complete

Description

Initiating actor: board member. Require the caller to be resolved to a principal with task.complete for this board; a role name supplied as business input is insufficient.

Trigger

The board member (ACT-BOARD-MEMBER) requests: complete task.

Context

Required caller context: resolved principal identifier, selected board identifier, authenticated status, granted case permissions, issuer, and validity status. The context originates from a trusted identity-and-permission authority configured for the application; claims in the task request are untrusted and cannot supply authorization.

The trust assumption is ASM-TASK-IDENTITY. Authentication is required; its mechanism and token/channel encoding belong to later design. Authorization is reevaluated for this invocation, and composed invocations would preserve the same principal and board context.

The actor is a role. The caller is the concrete principal for this request. Neither is an accountable human author for a Needs-to-L1 derivation.

Inputs

Name

Task identifier

Type: text

Meaning

Exact identifier to complete.

Required: true

Entry conditions

Requests may arrive with invalid or missing authorization context; the case's denial path remains defined. No downstream effect occurs before authorization succeeds.

Entities

Entity: ENT-TASK

Validations: VLD-TASK-ID, VLD-TASK-COMPLETE

Purpose

Use the canonical Task meaning and rules. Tools receive ready entities; no repeated validation of ready entity invariants is required.

Tools

Tool: TOL-TASK-STORE

Operation: OP-TASK-LOAD

Purpose

Use the declared capability and handle every stated observable outcome.

Tool: TOL-TASK-STORE

Operation: OP-TASK-COMPLETE

Purpose

Use the declared capability and handle every stated observable outcome.

Calls: none

Outcomes

`OUT-TASK-COMPLETE-DENIED`

Kind: refusal

Description

The caller context is absent, unresolved, expired, or lacks this case's permission for this board.

Effects

No tool call, no write, and no disclosure of task existence or content.

`OUT-TASK-COMPLETE-COMPLETED`

Kind: success

Description

Return the confirmed completed Task.

Effects

Status changes once; revision increases by one, title/identifier unchanged.

`OUT-TASK-COMPLETE-INVALID`

Kind: refusal

Description

Identifier is invalid.

Effects

No store access.

`OUT-TASK-COMPLETE-MISSING`

Kind: refusal

Description

Task absence was confirmed at read or write time.

Effects

This invocation makes no write.

OUT-TASK-COMPLETE-ALREADY

Kind: refusal

Description

Task was already completed.

Effects

No completion write.

OUT-TASK-COMPLETE-CONFLICT

Kind: refusal

Description

The Task changed after the observed read.

Effects

This invocation does not overwrite the competing change.

OUT-TASK-COMPLETE-STORAGE

Kind: failure

Description

Storage was unavailable or returned invalid stored state.

Effects

Read failures make no change. For a write, not_applied guarantees no change and unknown leaves the final state unresolved.

Functional requirements

RF-TASK-COMPLETE-AUTH

Statement

Resolve the initiating board member and require task.complete for the selected board in trusted caller context. Missing, unresolved, expired, or insufficient context is denied before any entity application or tool call. No anonymous execution is permitted. Origin: SRC-TASK-BRIEF.

Scope: case

Applies to: UC-TASK-COMPLETE

Flow references: N-TASK-COMPLETE-AUTH, N-TASK-COMPLETE-DENIED, E-TASK-COMPLETE-DENIED, E-TASK-COMPLETE-ALLOWED

RF-TASK-COMPLETE-ID

Statement

After authorization, apply VLD-TASK-ID to the supplied identifier; refuse invalid input before tool access.

Scope: case

Applies to: UC-TASK-COMPLETE

Flow references: N-TASK-COMPLETE-ID, N-TASK-COMPLETE-VALID, N-TASK-COMPLETE-INVALID, E-TASK-COMPLETE-2

RF-TASK-COMPLETE-LOAD

Statement

Read a ready Task and its observed revision with OP-TASK-LOAD. A confirmed NOT_FOUND is a refusal without write, distinct from unavailable or malformed storage.

Scope: operation

Applies to: UC-TASK-COMPLETE, OP-TASK-LOAD

Flow references: N-TASK-COMPLETE-LOAD, N-TASK-COMPLETE-MISSING, E-TASK-COMPLETE-3, E-TASK-COMPLETE-4

RF-TASK-COMPLETE-TRANSITION

Statement

Apply VLD-TASK-COMPLETE to the ready Task's current state. If already completed, refuse without writing. Otherwise produce a ready completed Task preserving identifier/title; do not repeat validation of its established invariants.

Scope: case

Applies to: UC-TASK-COMPLETE

Flow references: N-TASK-COMPLETE-TRANSITION, N-TASK-COMPLETE-ALLOWED, N-TASK-COMPLETE-ALREADY, E-TASK-COMPLETE-6, E-TASK-COMPLETE-8

RF-TASK-COMPLETE-WRITE

Statement

Call OP-TASK-COMPLETE with the completed Task and observed revision. Confirm only its successful result. NOT_FOUND or VERSION_CONFLICT refuses with no write by this operation; never overwrite a competing change.

Scope: operation

Applies to: UC-TASK-COMPLETE, OP-TASK-COMPLETE

Flow references: N-TASK-COMPLETE-MISSING, N-TASK-COMPLETE-STORE, N-TASK-COMPLETE-CONFLICT, N-TASK-COMPLETE-DONE, E-TASK-COMPLETE-9, E-TASK-COMPLETE-10, E-TASK-COMPLETE-11, E-TASK-COMPLETE-13

RF-TASK-COMPLETE-FAILURE

Statement

Propagate read unavailable/invalid-state failures without a write. On a write's STORE_UNAVAILABLE preserve not_applied versus unknown; do not claim completion or retry automatically.

Scope: operation

Applies to: UC-TASK-COMPLETE, OP-TASK-LOAD, OP-TASK-COMPLETE

Flow references: N-TASK-COMPLETE-ERROR, E-TASK-COMPLETE-5, E-TASK-COMPLETE-12

Nonfunctional requirements

RNF-TASK-COMPLETE-ATOMICITY

Scope: operation

Applies to: UC-TASK-COMPLETE, OP-TASK-COMPLETE

Statement

Concurrent completion must not lose an update or report two writes for the same observed revision. Origin: SRC-TASK-BRIEF (synthetic teaching brief).

Verifiable criterion

For two conditional writes using the same observed revision, at most one is confirmed; the other returns conflict or an accurately classified failure.

Verification

Coordinate concurrent attempts at the conditional-write boundary and inspect confirmed revisions and outcomes.

References: SRC-TASK-BRIEF

Applicability

Applies under the explicitly stated case and tool conditions. This is an obligation for a future implementation, not a measured claim about this repository.

Flows

Main: GRAPH-TASK-COMPLETE

Subflows: none

Flow text

Authorize trusted caller context first (RF-TASK-COMPLETE-AUTH). Validate the identifier, then load a ready Task (RF-TASK-COMPLETE-ID, RF-TASK-COMPLETE-LOAD).

Refuse a confirmed absence; report a read failure without attempting a write. Apply the entity's transition to the ready Task; refuse already-completed state (RF-TASK-COMPLETE-TRANSITION).

Conditionally retain the completed Task with its observed revision. Distinguish success, concurrent conflict, confirmed absence, and unavailable-write outcome (RF-TASK-COMPLETE-WRITE, RF-TASK-COMPLETE-FAILURE).

Acceptance criteria

AC-TASK-COMPLETE-AUTH

Statement

Try absent context, an unresolved identity, an expired context, and a member lacking this permission. Every attempt is denied before tool invocation, with no task state or existence disclosed.

Verifies: RF-TASK-COMPLETE-AUTH

AC-TASK-COMPLETE-SUCCESS

Statement

For TASK001 open at revision zero, completion returns the same identifier/title, completed status and revision one; later listing reflects that state.

Verifies: RF-TASK-COMPLETE-TRANSITION, RF-TASK-COMPLETE-WRITE

AC-TASK-COMPLETE-MISSING

Statement

A confirmed absent identifier produces not-found without calling OP-TASK-COMPLETE.

Verifies: RF-TASK-COMPLETE-LOAD

AC-TASK-COMPLETE-INVALID

Statement

An empty or whitespace-only identifier is refused before load or completion tools are invoked.

Verifies: RF-TASK-COMPLETE-ID

AC-TASK-COMPLETE-ALREADY

Statement

A ready completed Task produces already-completed refusal and no write; its revision remains unchanged.

Verifies: RF-TASK-COMPLETE-TRANSITION

AC-TASK-COMPLETE-CONFLICT

Statement

Change the retained revision between load and completion. Return VERSION_CONFLICT, preserving the competing state.

Verifies: RF-TASK-COMPLETE-WRITE, RNF-TASK-COMPLETE-ATOMICITY

AC-TASK-COMPLETE-UNKNOWN

Statement

Lose the completion confirmation after issuing a write. Return unknown without success/no-change claims or automatic retry.

Verifies: RF-TASK-COMPLETE-FAILURE

Traceability

TRC-TASK-COMPLETE-ORIGIN

Source: SRC-TASK-BRIEF

Targets: RF-TASK-COMPLETE-AUTH, RF-TASK-COMPLETE-ID, RF-TASK-COMPLETE-LOAD, RF-TASK-COMPLETE-TRANSITION, RF-TASK-COMPLETE-WRITE, RF-TASK-COMPLETE-FAILURE, RNF-TASK-COMPLETE-ATOMICITY

Note

Synthetic requirements selected for this teaching example; not evidence of stakeholder approval.

TRC-TASK-COMPLETE-AC-1

Source: AC-TASK-COMPLETE-AUTH

Targets: RF-TASK-COMPLETE-AUTH

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-COMPLETE-AC-2

Source: AC-TASK-COMPLETE-SUCCESS

Targets: RF-TASK-COMPLETE-TRANSITION, RF-TASK-COMPLETE-WRITE

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-COMPLETE-AC-3

Source: AC-TASK-COMPLETE-MISSING

Targets: RF-TASK-COMPLETE-LOAD

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-COMPLETE-AC-4

Source: AC-TASK-COMPLETE-INVALID

Targets: RF-TASK-COMPLETE-ID

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-COMPLETE-AC-5

Source: AC-TASK-COMPLETE-ALREADY

Targets: RF-TASK-COMPLETE-TRANSITION

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-COMPLETE-AC-6

Source: AC-TASK-COMPLETE-CONFLICT

Targets: RF-TASK-COMPLETE-WRITE, RNF-TASK-COMPLETE-ATOMICITY

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-COMPLETE-AC-7

Source: AC-TASK-COMPLETE-UNKNOWN

Targets: RF-TASK-COMPLETE-FAILURE

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

Assumptions: ASM-TASK-SINGLE-BOARD, ASM-TASK-IDENTITY

Sources: SRC-TASK-BRIEF, SRC-TASK-OAL

Open issues: none

USE CASE UC-TASK-LIST

Name

List tasks

Objective

Observe all current Tasks in the authorized board without changing them.

Actors

Actor: ACT-BOARD-MEMBER

Permission

task.list

Description

Initiating actor: board member. Require the caller to be resolved to a principal with task.list for this board; a role name supplied as business input is insufficient.

Trigger

The board member (ACT-BOARD-MEMBER) requests: list tasks.

Context

Required caller context: resolved principal identifier, selected board identifier, authenticated status, granted case permissions, issuer, and validity status. The context originates from a trusted identity-and-permission authority configured for the application; claims in the task request are untrusted and cannot supply authorization.

The trust assumption is ASM-TASK-IDENTITY. Authentication is required; its mechanism and token/channel encoding belong to later design. Authorization is reevaluated for this invocation, and composed invocations would preserve the same principal and board context.

The actor is a role. The caller is the concrete principal for this request. Neither is an accountable human author for a Needs-to-L1 derivation.

Inputs: none

Entry conditions

Requests may arrive with invalid or missing authorization context; the case's denial path remains defined. No downstream effect occurs before authorization succeeds.

Entities

Entity: ENT-TASK

Validations: none

Purpose

Use the canonical Task meaning and rules. Tools receive ready entities; no repeated validation of ready entity invariants is required.

Tools

Tool: TOL-TASK-STORE

Operation: OP-TASK-LIST

Purpose

Use the declared capability and handle every stated observable outcome.

Calls: none

Outcomes

OUT-TASK-LIST-DENIED

Kind: refusal

Description

The caller context is absent, unresolved, expired, or lacks this case's permission for this board.

Effects

No tool call, no write, and no disclosure of task existence or content.

OUT-TASK-LIST-LISTED

Kind: success

Description

Return the complete ordered snapshot, including an empty list for a confirmed empty board.

Effects

No task changes.

OUT-TASK-LIST-STORAGE

Kind: failure

Description

The requested snapshot could not be obtained or contained invalid stored state.

Effects

No task changes; no usable complete list is claimed.

Functional requirements

RF-TASK-LIST-AUTH

Statement

Resolve the initiating board member and require task.list for the selected board in trusted caller context. Missing, unresolved, expired, or insufficient context is denied before any entity application or tool call. No anonymous execution is permitted. Origin: SRC-TASK-BRIEF.

Scope: case

Applies to: UC-TASK-LIST

Flow references: N-TASK-LIST-AUTH, N-TASK-LIST-DENIED, E-TASK-LIST-DENIED, E-TASK-LIST-ALLOWED

RF-TASK-LIST-SNAPSHOT

Statement

After authorization, call OP-TASK-LIST for the trusted board. Return every ready Task exactly once in the declared identifier order, preserving title, state, and revision. A confirmed empty snapshot is a successful empty list. Do not revalidate ready entities.

Scope: case

Applies to: UC-TASK-LIST, OP-TASK-LIST

Flow references: N-TASK-LIST-READ, N-TASK-LIST-RETURN, E-TASK-LIST-2

RF-TASK-LIST-FAILURE

Statement

Report unavailable or invalid-state retrieval as a failure, preserving its distinction. Do not return an empty or partial success and make no task changes.

Scope: operation

Applies to: UC-TASK-LIST, OP-TASK-LIST

Flow references: N-TASK-LIST-FAIL, E-TASK-LIST-1

Nonfunctional requirements

RNF-TASK-LIST-CONSISTENCY

Scope: operation

Applies to: UC-TASK-LIST, OP-TASK-LIST

Statement

The list represents one consistent snapshot of the authorized board. Origin: SRC-TASK-BRIEF (synthetic teaching brief).

Verifiable criterion

Every returned Task belongs to the same board snapshot, appears exactly once, and has mutually consistent fields/revision; no read changes a Task.

Verification

Exercise listing alongside concurrent changes, verify against the chosen store's snapshot guarantee, and inspect before/after write counters.

References: SRC-TASK-BRIEF

Applicability

Applies under the explicitly stated case and tool conditions. This is an obligation for a future implementation, not a measured claim about this repository.

RNF-TASK-LIST-RETENTION

Reference to: RNF-TASK-CREATE-RETENTION

Scope: system

Applies to: UC-TASK-LIST, OP-TASK-LIST

References: SRC-TASK-BRIEF

Applicability

The canonical confirmed-retention obligation defined in creation also applies when listing after a normal store restart.

Flows

Main: GRAPH-TASK-LIST

Subflows: none

Flow text

Resolve and authorize caller context before retrieving any Task (RF-TASK-LIST-AUTH).

Read one complete snapshot with OP-TASK-LIST and return the ready Tasks unchanged. An empty successful snapshot means confirmed emptiness (RF-TASK-LIST-SNAPSHOT).

Report retrieval failure as failure, never as an empty or partial successful list (RF-TASK-LIST-FAILURE).

Acceptance criteria

AC-TASK-LIST-AUTH

Statement

Try absent context, an unresolved identity, an expired context, and a member lacking this permission. Every attempt is denied before tool invocation, with no task state or existence disclosed.

Verifies: RF-TASK-LIST-AUTH

AC-TASK-LIST-CONTENT

Statement

Given open TASK002 and completed TASK001, return TASK001 then TASK002 with exact fields, once each, with no state or revision changed.

Verifies: RF-TASK-LIST-SNAPSHOT, RNF-TASK-LIST-CONSISTENCY

AC-TASK-LIST-EMPTY

Statement

A successful read of an empty board returns an empty list; an unavailable store instead returns failure.

Verifies: RF-TASK-LIST-SNAPSHOT, RF-TASK-LIST-FAILURE

AC-TASK-LIST-INTEGRITY

Statement

If hydration finds an invalid stored Task, report STORE_INVALID_STATE without publishing a partial successful list.

Verifies: RF-TASK-LIST-FAILURE

Traceability

TRC-TASK-LIST-ORIGIN

Source: SRC-TASK-BRIEF

Targets: RF-TASK-LIST-AUTH, RF-TASK-LIST-SNAPSHOT, RF-TASK-LIST-FAILURE, RNF-TASK-LIST-CONSISTENCY, RNF-TASK-LIST-RETENTION

Note

Synthetic requirements selected for this teaching example; not evidence of stakeholder approval.

TRC-TASK-LIST-AC-1

Source: AC-TASK-LIST-AUTH

Targets: RF-TASK-LIST-AUTH

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-LIST-AC-2

Source: AC-TASK-LIST-CONTENT

Targets: RF-TASK-LIST-SNAPSHOT, RNF-TASK-LIST-CONSISTENCY

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-LIST-AC-3

Source: AC-TASK-LIST-EMPTY

Targets: RF-TASK-LIST-SNAPSHOT, RF-TASK-LIST-FAILURE

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-LIST-AC-4

Source: AC-TASK-LIST-INTEGRITY

Targets: RF-TASK-LIST-FAILURE

Note

Acceptance criterion states an observable obligation, not a claim that a task application has been tested.

TRC-TASK-LIST-RETENTION

Source: RNF-TASK-LIST-RETENTION

Targets: RNF-TASK-CREATE-RETENTION, AC-TASK-CREATE-SUCCESS

Note

One canonical integration criterion covers confirmed creation and subsequent listing after a normal restart; no duplicate full requirement is authored.

Assumptions: ASM-TASK-SINGLE-BOARD, ASM-TASK-IDENTITY

Sources: SRC-TASK-BRIEF, SRC-TASK-OAL

Open issues: none

Illustrative interaction mapping

The mappings below show which inputs/results cross the interaction boundary. They are explanatory examples, not a choice of an L4 channel.

BOUNDARY BND-TASK-ILLUSTRATION

Name

Channel-independent interaction illustration

Modality: other

Purpose

Origin: SRC-TASK-BRIEF. This is an illustrative mapping of domain inputs/outcomes. A later solution may select an L4 channel; such a selection must not retroactively become a stakeholder L1 requirement.

Audience

Authorized board members.

Interactions

INT-TASK-CREATE

Name

Create task

Description

Illustrative interaction mapping only. No web route, screen, command syntax, or protocol is a normative L1 requirement.

Use case: UC-TASK-CREATE

Trigger

The board member (ACT-BOARD-MEMBER) requests: create task.

Input mapping

Source: request.task_identifier

Target: UC-TASK-CREATE.input.task_identifier

Meaning

Requested stable task identifier.

Source: request.title

Target: UC-TASK-CREATE.input.title

Meaning

The work to remember.

Result mapping

Source: OUT-TASK-CREATE-DENIED

Target: result.out-task-create-denied

Meaning

The caller context is absent, unresolved, expired, or lacks this case's permission for this board.

Source: OUT-TASK-CREATE-CREATED

Target: result.out-task-create-created

Meaning

The retained Task is returned.

Source: OUT-TASK-CREATE-INVALID

Target: result.out-task-create-invalid

Meaning

The new-task validity rule failed.

Source: OUT-TASK-CREATE-DUPLICATE

Target: result.out-task-create-duplicate

Meaning

The identifier was already retained.

Source: OUT-TASK-CREATE-UNAVAILABLE

Target: result.out-task-create-unavailable

Meaning

Storage did not confirm creation; report write_outcome.

Requirement refs: RF-TASK-CREATE-AUTH, RF-TASK-CREATE-DATA, RF-TASK-CREATE-RETAIN, RF-TASK-CREATE-DUPLICATE, RF-TASK-CREATE-FAILURE

Resources: none

Examples: none

INT-TASK-COMPLETE

Name

Complete task

Description

Illustrative interaction mapping only. No web route, screen, command syntax, or protocol is a normative L1 requirement.

Use case: UC-TASK-COMPLETE

Trigger

The board member (ACT-BOARD-MEMBER) requests: complete task.

Input mapping

Source: request.task_identifier

Target: UC-TASK-COMPLETE.input.task_identifier

Meaning

Exact identifier to complete.

Result mapping

Source: OUT-TASK-COMPLETE-DENIED

Target: result.out-task-complete-denied

Meaning

The caller context is absent, unresolved, expired, or lacks this case's permission for this board.

Source: OUT-TASK-COMPLETE-COMPLETED

Target: result.out-task-complete-completed

Meaning

Return the confirmed completed Task.

Source: OUT-TASK-COMPLETE-INVALID

Target: result.out-task-complete-invalid

Meaning

Identifier is invalid.

Source: OUT-TASK-COMPLETE-MISSING

Target: result.out-task-complete-missing

Meaning

Task absence was confirmed at read or write time.

Source: OUT-TASK-COMPLETE-ALREADY

Target: result.out-task-complete-already

Meaning

Task was already completed.

Source: OUT-TASK-COMPLETE-CONFLICT

Target: result.out-task-complete-conflict

Meaning

The Task changed after the observed read.

Source: OUT-TASK-COMPLETE-STORAGE

Target: result.out-task-complete-storage

Meaning

Storage was unavailable or returned invalid stored state.

Requirement refs: RF-TASK-COMPLETE-AUTH, RF-TASK-COMPLETE-ID, RF-TASK-COMPLETE-LOAD, RF-TASK-COMPLETE-TRANSITION, RF-TASK-COMPLETE-WRITE, RF-TASK-COMPLETE-FAILURE

Resources: none

Examples: none

`INT-TASK-LIST`

Name

List tasks

Description

Illustrative interaction mapping only. No web route, screen, command syntax, or protocol is a normative L1 requirement.

Use case: UC-TASK-LIST

Trigger

The board member (ACT-BOARD-MEMBER) requests: list tasks.

Input mapping: none

Result mapping

Source: OUT-TASK-LIST-DENIED

Target: result.out-task-list-denied

Meaning

The caller context is absent, unresolved, expired, or lacks this case's permission for this board.

Source: OUT-TASK-LIST-LISTED

Target: result.out-task-list-listed

Meaning

Return the complete ordered snapshot, including an empty list for a confirmed empty board.

Source: OUT-TASK-LIST-STORAGE

Target: result.out-task-list-storage

Meaning

The requested snapshot could not be obtained or contained invalid stored state.

Requirement refs: RF-TASK-LIST-AUTH, RF-TASK-LIST-SNAPSHOT, RF-TASK-LIST-FAILURE

Resources: none

Examples: none

Assumptions: none

Sources: SRC-TASK-BRIEF

Open issues: none

Appendix: graphs

Graph GRAPH-VLD-TASK-ID

Task identifier

Kind: flow

Nodes

Node N-VLD-TASK-ID-START (start)

Receive candidate

Node N-VLD-TASK-ID-CHECK (decision)

All conditions hold?

Node N-VLD-TASK-ID-PASS (result)

Pass unchanged

Node N-VLD-TASK-ID-FAIL (refusal)

Fail; no mutation

Edges

Edge E-VLD-TASK-ID-1: N-VLD-TASK-ID-START → N-VLD-TASK-ID-CHECK

Requirement refs: none

Edge E-VLD-TASK-ID-2: N-VLD-TASK-ID-CHECK → N-VLD-TASK-ID-PASS

Label

yes

Requirement refs: none

Edge E-VLD-TASK-ID-3: N-VLD-TASK-ID-CHECK → N-VLD-TASK-ID-FAIL

Label

no

Requirement refs: none

Notes

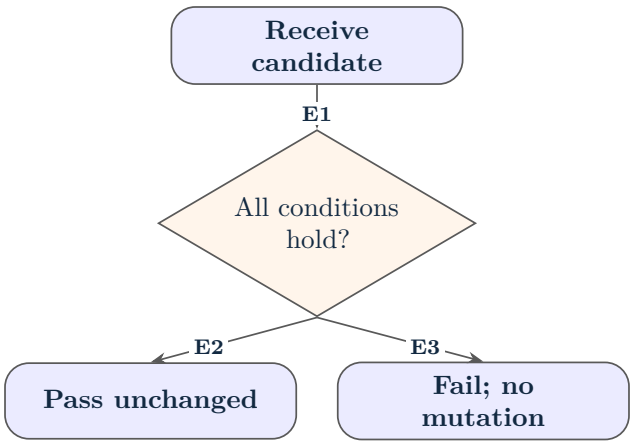
The entity owns these conditions. Applying them needs no collaborator.

Topology of GRAPH-VLD-TASK-ID:
Nodes:
- N-VLD-TASK-ID-START (start)
- N-VLD-TASK-ID-CHECK (decision)
- N-VLD-TASK-ID-PASS (result)
- N-VLD-TASK-ID-FAIL (refusal)

Edges:
- E-VLD-TASK-ID-1: N-VLD-TASK-ID-START -> N-VLD-TASK-ID-CHECK
- E-VLD-TASK-ID-2: N-VLD-TASK-ID-CHECK -> N-VLD-TASK-ID-PASS
- E-VLD-TASK-ID-3: N-VLD-TASK-ID-CHECK -> N-VLD-TASK-ID-FAIL

Task identifier

GRAPH-VLD-TASK-ID



Edge key

E1 E-VLD-TASK-ID-1
E2 E-VLD-TASK-ID-2 — yes
E3 E-VLD-TASK-ID-3 — no

Graph `GRAPH-VLD-TASK-NEW`

New task validity

Kind: flow

Nodes

Node `N-VLD-TASK-NEW-START` (start)

Receive candidate

Node `N-VLD-TASK-NEW-CHECK` (decision)

All conditions hold?

Node `N-VLD-TASK-NEW-PASS` (result)

Pass unchanged

Node `N-VLD-TASK-NEW-FAIL` (refusal)

Fail; no mutation

Edges

Edge `E-VLD-TASK-NEW-1: N-VLD-TASK-NEW-START → N-VLD-TASK-NEW-CHECK`

Requirement refs: none

Edge `E-VLD-TASK-NEW-2: N-VLD-TASK-NEW-CHECK → N-VLD-TASK-NEW-PASS`

Label

yes

Requirement refs: none

Edge `E-VLD-TASK-NEW-3: N-VLD-TASK-NEW-CHECK → N-VLD-TASK-NEW-FAIL`

Label

no

Requirement refs: none

Notes

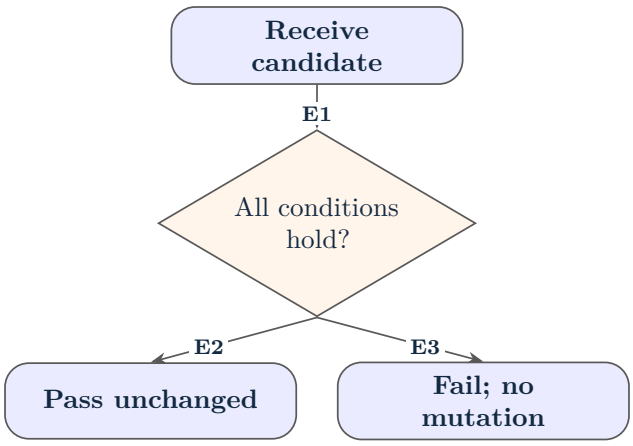
The entity owns these conditions. Applying them needs no collaborator.

Topology of GRAPH-VLD-TASK-NEW:
Nodes:
- N-VLD-TASK-NEW-START (start)
- N-VLD-TASK-NEW-CHECK (decision)
- N-VLD-TASK-NEW-PASS (result)
- N-VLD-TASK-NEW-FAIL (refusal)

Edges:
- E-VLD-TASK-NEW-1: N-VLD-TASK-NEW-START -> N-VLD-TASK-NEW-CHECK
- E-VLD-TASK-NEW-2: N-VLD-TASK-NEW-CHECK -> N-VLD-TASK-NEW-PASS
- E-VLD-TASK-NEW-3: N-VLD-TASK-NEW-CHECK -> N-VLD-TASK-NEW-FAIL

New task validity

GRAPH-VLD-TASK-NEW



Edge key

E1 E-VLD-TASK-NEW-1
E2 E-VLD-TASK-NEW-2 — yes
E3 E-VLD-TASK-NEW-3 — no

Graph GRAPH-VLD-TASK-COMPLETE

Completion transition

Kind: flow

Nodes

Node N-VLD-TASK-COMPLETE-START (start)

Receive candidate

Node N-VLD-TASK-COMPLETE-CHECK (decision)

All conditions hold?

Node N-VLD-TASK-COMPLETE-PASS (result)

Pass unchanged

Node N-VLD-TASK-COMPLETE-FAIL (refusal)

Fail; no mutation

Edges

Edge E-VLD-TASK-COMPLETE-1: N-VLD-TASK-COMPLETE-START → N-VLD-TASK-COMPLETE-CHECK

Requirement refs: none

Edge E-VLD-TASK-COMPLETE-2: N-VLD-TASK-COMPLETE-CHECK → N-VLD-TASK-COMPLETE-PASS

Label

yes

Requirement refs: none

Edge E-VLD-TASK-COMPLETE-3: N-VLD-TASK-COMPLETE-CHECK → N-VLD-TASK-COMPLETE-FAIL

Label

no

Requirement refs: none

Notes

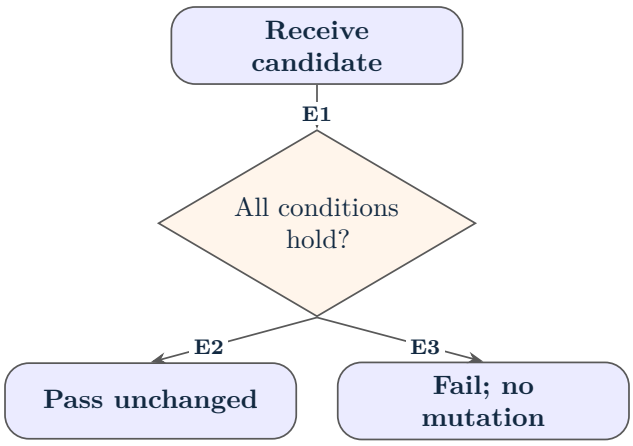
The entity owns these conditions. Applying them needs no collaborator.

Topology of GRAPH-VLD-TASK-COMPLETE:
Nodes:
- N-VLD-TASK-COMPLETE-START (start)
- N-VLD-TASK-COMPLETE-CHECK (decision)
- N-VLD-TASK-COMPLETE-PASS (result)
- N-VLD-TASK-COMPLETE-FAIL (refusal)

Edges:
- E-VLD-TASK-COMPLETE-1: N-VLD-TASK-COMPLETE-START -> N-VLD-TASK-COMPLETE-CHECK
- E-VLD-TASK-COMPLETE-2: N-VLD-TASK-COMPLETE-CHECK -> N-VLD-TASK-COMPLETE-PASS
- E-VLD-TASK-COMPLETE-3: N-VLD-TASK-COMPLETE-CHECK -> N-VLD-TASK-COMPLETE-FAIL

Completion transition

GRAPH-VLD-TASK-COMPLETE



Edge key

E1 E-VLD-TASK-COMPLETE-1
E2 E-VLD-TASK-COMPLETE-2 — yes
E3 E-VLD-TASK-COMPLETE-3 — no

Graph GRAPH-TASK-CREATE

Create task

Kind: flow

Nodes

Node N-TASK-CREATE-START (start)

Receive request

Node N-TASK-CREATE-AUTH (decision)

Trusted permission?

Requirement refs: RF-TASK-CREATE-AUTH

Node N-TASK-CREATE-DENIED (refusal)

Denied; no tool call

Requirement refs: RF-TASK-CREATE-AUTH

Node N-TASK-CREATE-VALIDATE (validation)

Apply new-task validity

Call: validation VLD-TASK-NEW

Requirement refs: RF-TASK-CREATE-DATA

Node N-TASK-CREATE-VALID (decision)

Candidate valid?

Requirement refs: RF-TASK-CREATE-DATA

Node N-TASK-CREATE-INVALID (refusal)

Invalid; no write

Requirement refs: RF-TASK-CREATE-DATA

Node N-TASK-CREATE-STORE (operation)

Retain ready Task

Call: operation OP-TASK-CREATE

Requirement refs: RF-TASK-CREATE-RETAIN

Node N-TASK-CREATE-DUPLICATE (refusal)

Duplicate; no change

Requirement refs: RF-TASK-CREATE-DUPLICATE

Node N-TASK-CREATE-UNAVAILABLE (failure)

Report write outcome

Requirement refs: RF-TASK-CREATE-FAILURE

Node N-TASK-CREATE-CREATED (result)

Created at revision 0

Requirement refs: RF-TASK-CREATE-RETAIN

Edges

Edge E-TASK-CREATE-START: N-TASK-CREATE-START → N-TASK-CREATE-AUTH

Requirement refs: none

Edge E-TASK-CREATE-DENIED: N-TASK-CREATE-AUTH → N-TASK-CREATE-DENIED

Label

missing / unresolved / insufficient

Requirement refs: RF-TASK-CREATE-AUTH

Edge E-TASK-CREATE-ALLOWED: N-TASK-CREATE-AUTH → N-TASK-CREATE-VALIDATE

Label

authorized

Requirement refs: RF-TASK-CREATE-AUTH

Edge E-TASK-CREATE-1: N-TASK-CREATE-VALIDATE → N-TASK-CREATE-VALID

Requirement refs: none

Edge E-TASK-CREATE-2: N-TASK-CREATE-VALID → N-TASK-CREATE-INVALID

Label

fail

Requirement refs: RF-TASK-CREATE-DATA

Edge E-TASK-CREATE-3: N-TASK-CREATE-VALID → N-TASK-CREATE-STORE

Label

pass

Requirement refs: RF-TASK-CREATE-RETAIN

Edge E-TASK-CREATE-4: N-TASK-CREATE-STORE → N-TASK-CREATE-DUPLICATE

Label

DUPLICATE_ID

Requirement refs: RF-TASK-CREATE-DUPLICATE

Edge E-TASK-CREATE-5: N-TASK-CREATE-STORE → N-TASK-CREATE-UNAVAILABLE

Label

STORE_UNAVAILABLE

Requirement refs: RF-TASK-CREATE-FAILURE

Edge E-TASK-CREATE-6: N-TASK-CREATE-STORE → N-TASK-CREATE-CREATED

Label

confirmed

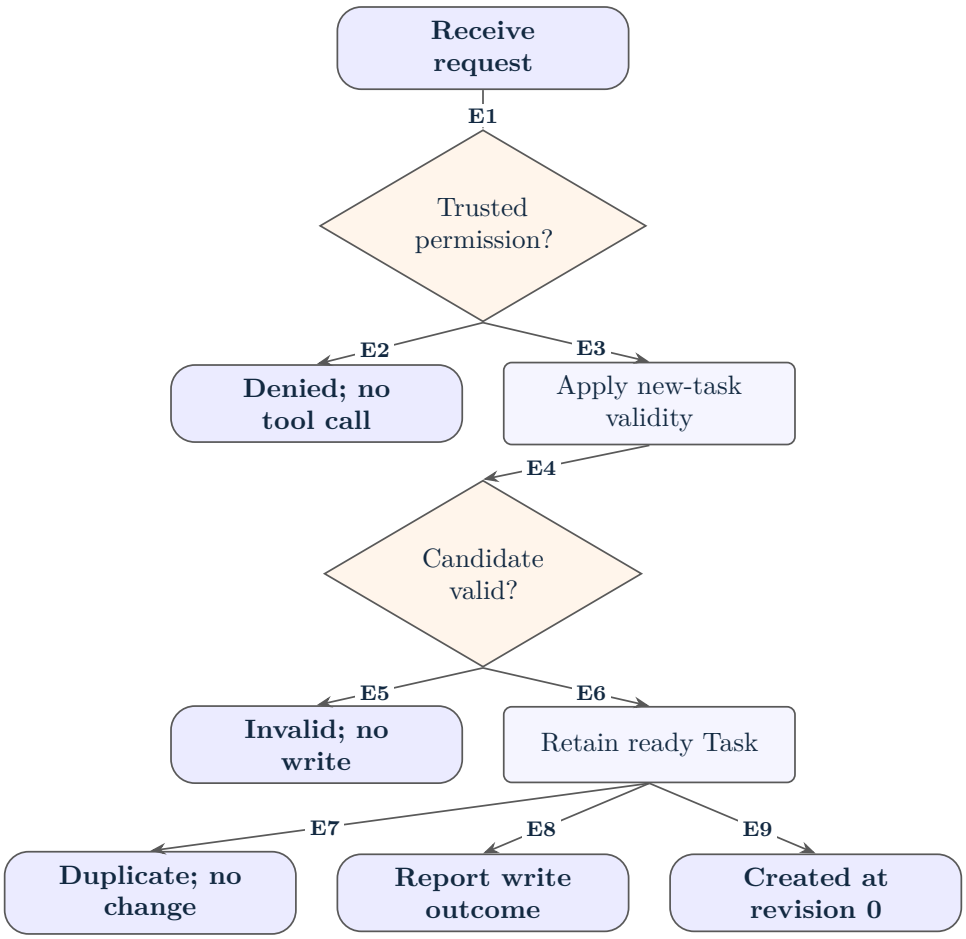
Requirement refs: RF-TASK-CREATE-RETAIN

Topology of GRAPH-TASK-CREATE:
Nodes:
- N-TASK-CREATE-START (start)
- N-TASK-CREATE-AUTH (decision)
- N-TASK-CREATE-DENIED (refusal)
- N-TASK-CREATE-VALIDATE (validation)
- N-TASK-CREATE-VALID (decision)
- N-TASK-CREATE-INVALID (refusal)
- N-TASK-CREATE-STORE (operation)
- N-TASK-CREATE-DUPLICATE (refusal)
- N-TASK-CREATE-UNAVAILABLE (failure)
- N-TASK-CREATE-CREATED (result)

Edges:
- E-TASK-CREATE-START: N-TASK-CREATE-START -> N-TASK-CREATE-AUTH
- E-TASK-CREATE-DENIED: N-TASK-CREATE-AUTH -> N-TASK-CREATE-DENIED
- E-TASK-CREATE-ALLOWED: N-TASK-CREATE-AUTH -> N-TASK-CREATE-VALIDATE
- E-TASK-CREATE-1: N-TASK-CREATE-VALIDATE -> N-TASK-CREATE-VALID
- E-TASK-CREATE-2: N-TASK-CREATE-VALID -> N-TASK-CREATE-INVALID
- E-TASK-CREATE-3: N-TASK-CREATE-VALID -> N-TASK-CREATE-STORE
- E-TASK-CREATE-4: N-TASK-CREATE-STORE -> N-TASK-CREATE-DUPLICATE
- E-TASK-CREATE-5: N-TASK-CREATE-STORE -> N-TASK-CREATE-UNAVAILABLE
- E-TASK-CREATE-6: N-TASK-CREATE-STORE -> N-TASK-CREATE-CREATED

Create task

GRAPH-TASK-CREATE



Edge key

E1 E-TASK-CREATE-START
E2 E-TASK-CREATE-DENIED — missing / unresolved / insufficient
E3 E-TASK-CREATE-ALLOWED — authorized
E4 E-TASK-CREATE-1
E5 E-TASK-CREATE-2 — fail
E6 E-TASK-CREATE-3 — pass
E7 E-TASK-CREATE-4 — DUPLICATE_ID
E8 E-TASK-CREATE-5 — STORE_UNAVAILABLE
E9 E-TASK-CREATE-6 — confirmed

Graph GRAPH-TASK-COMPLETE

Complete task

Kind: flow

Nodes

Node N-TASK-COMPLETE-START (start)

Receive request

Node N-TASK-COMPLETE-AUTH (decision)

Trusted permission?

Requirement refs: RF-TASK-COMPLETE-AUTH

Node N-TASK-COMPLETE-DENIED (refusal)

Denied; no tool call

Requirement refs: RF-TASK-COMPLETE-AUTH

Node N-TASK-COMPLETE-ID (validation)

Apply identifier validity

Call: validation VLD-TASK-ID

Requirement refs: RF-TASK-COMPLETE-ID

Node N-TASK-COMPLETE-VALID (decision)

Identifier valid?

Requirement refs: RF-TASK-COMPLETE-ID

Node N-TASK-COMPLETE-INVALID (refusal)

Invalid identifier

Requirement refs: RF-TASK-COMPLETE-ID

Node N-TASK-COMPLETE-LOAD (operation)

Load ready Task

Call: operation OP-TASK-LOAD

Requirement refs: RF-TASK-COMPLETE-LOAD

Node N-TASK-COMPLETE-MISSING (refusal)

Task not found

Requirement refs: RF-TASK-COMPLETE-LOAD, RF-TASK-COMPLETE-WRITE

Node N-TASK-COMPLETE-TRANSITION (validation)

Apply completion rule

Call: validation VLD-TASK-COMPLETE

Requirement refs: RF-TASK-COMPLETE-TRANSITION

Node N-TASK-COMPLETE-ALLOWED (decision)

Transition allowed?

Requirement refs: RF-TASK-COMPLETE-TRANSITION

Node N-TASK-COMPLETE-ALREADY (refusal)

Already completed

Requirement refs: RF-TASK-COMPLETE-TRANSITION

Node N-TASK-COMPLETE-STORE (operation)

Conditional completion

Call: operation OP-TASK-COMPLETE

Requirement refs: RF-TASK-COMPLETE-WRITE

Node N-TASK-COMPLETE-CONFLICT (refusal)

Version conflict

Requirement refs: RF-TASK-COMPLETE-WRITE

Node N-TASK-COMPLETE-ERROR (failure)

Report storage outcome

Requirement refs: RF-TASK-COMPLETE-FAILURE

Node N-TASK-COMPLETE-DONE (result)

Completion confirmed

Requirement refs: RF-TASK-COMPLETE-WRITE

Edges

Edge E-TASK-COMPLETE-START: N-TASK-COMPLETE-START → N-TASK-COMPLETE-AUTH

Requirement refs: none

Edge E-TASK-COMPLETE-DENIED: N-TASK-COMPLETE-AUTH → N-TASK-COMPLETE-DENIED

Label

missing / unresolved / insufficient

Requirement refs: RF-TASK-COMPLETE-AUTH

Edge E-TASK-COMPLETE-ALLOWED: N-TASK-COMPLETE-AUTH → N-TASK-COMPLETE-ID

Label

authorized

Requirement refs: RF-TASK-COMPLETE-AUTH

Edge E-TASK-COMPLETE-1: N-TASK-COMPLETE-ID → N-TASK-COMPLETE-VALID

Requirement refs: none

Edge E-TASK-COMPLETE-2: N-TASK-COMPLETE-VALID → N-TASK-COMPLETE-INVALID

Label

fail

Requirement refs: RF-TASK-COMPLETE-ID

Edge E-TASK-COMPLETE-3: N-TASK-COMPLETE-VALID → N-TASK-COMPLETE-LOAD

Label

pass

Requirement refs: RF-TASK-COMPLETE-LOAD

Edge E-TASK-COMPLETE-4: N-TASK-COMPLETE-LOAD → N-TASK-COMPLETE-MISSING

Label

NOT_FOUND

Requirement refs: RF-TASK-COMPLETE-LOAD

Edge E-TASK-COMPLETE-5: N-TASK-COMPLETE-LOAD → N-TASK-COMPLETE-ERROR

Label

unavailable / invalid state

Requirement refs: RF-TASK-COMPLETE-FAILURE

Edge E-TASK-COMPLETE-6: N-TASK-COMPLETE-LOAD → N-TASK-COMPLETE-TRANSITION

Label

ready Task

Requirement refs: RF-TASK-COMPLETE-TRANSITION

Edge E-TASK-COMPLETE-7: N-TASK-COMPLETE-TRANSITION → N-TASK-COMPLETE-ALLOWED

Requirement refs: none

Edge E-TASK-COMPLETE-8: N-TASK-COMPLETE-ALLOWED → N-TASK-COMPLETE-ALREADY

Label

fail

Requirement refs: RF-TASK-COMPLETE-TRANSITION

Edge E-TASK-COMPLETE-9: N-TASK-COMPLETE-ALLOWED → N-TASK-COMPLETE-STORE

Label

pass

Requirement refs: RF-TASK-COMPLETE-WRITE

Edge E-TASK-COMPLETE-10: N-TASK-COMPLETE-STORE → N-TASK-COMPLETE-CONFLICT

Label

VERSION_CONFLICT

Requirement refs: RF-TASK-COMPLETE-WRITE

Edge E-TASK-COMPLETE-11: N-TASK-COMPLETE-STORE → N-TASK-COMPLETE-MISSING

Label

NOT_FOUND

Requirement refs: RF-TASK-COMPLETE-WRITE

Edge E-TASK-COMPLETE-12: N-TASK-COMPLETE-STORE → N-TASK-COMPLETE-ERROR

Label

STORE_UNAVAILABLE

Requirement refs: RF-TASK-COMPLETE-FAILURE

Edge E-TASK-COMPLETE-13: N-TASK-COMPLETE-STORE → N-TASK-COMPLETE-DONE

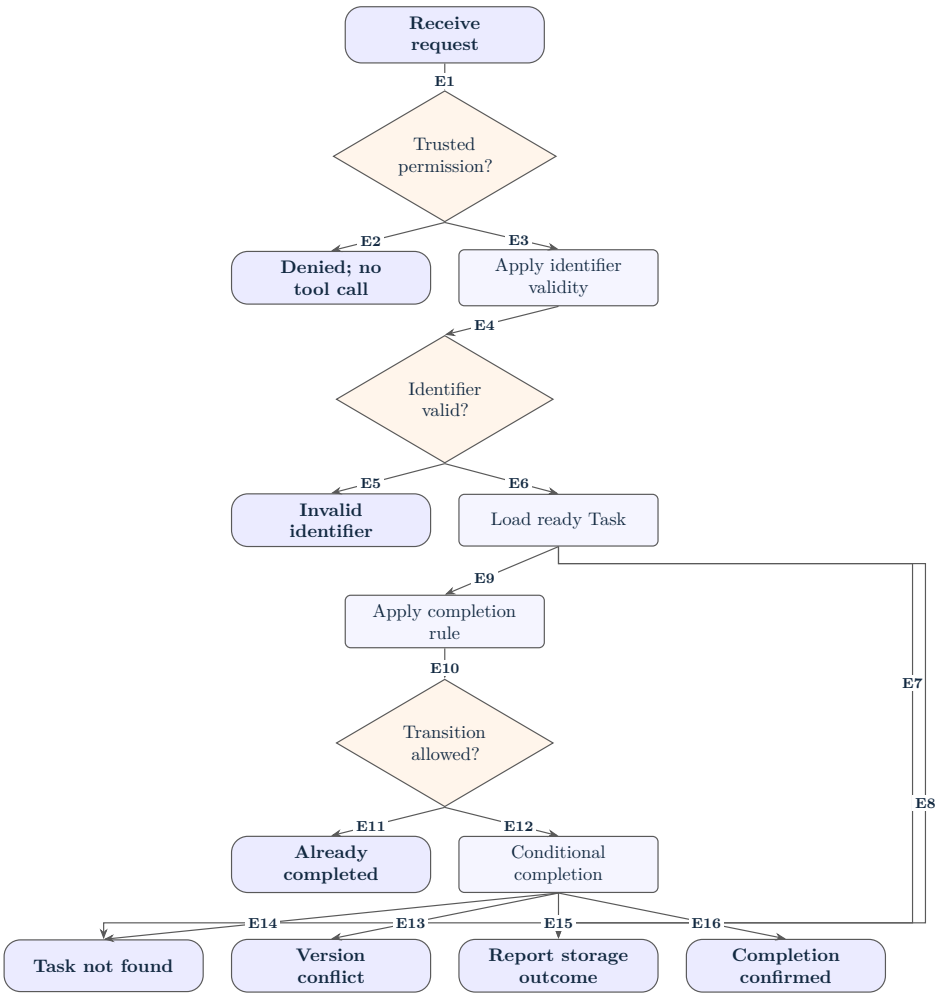
Label

confirmed

Requirement refs: RF-TASK-COMPLETE-WRITE
Topology of GRAPH-TASK-COMPLETE:
Nodes:
- N-TASK-COMPLETE-START (start)
- N-TASK-COMPLETE-AUTH (decision)
- N-TASK-COMPLETE-DENIED (refusal)
- N-TASK-COMPLETE-ID (validation)
- N-TASK-COMPLETE-VALID (decision)
- N-TASK-COMPLETE-INVALID (refusal)
- N-TASK-COMPLETE-LOAD (operation)
- N-TASK-COMPLETE-MISSING (refusal)
- N-TASK-COMPLETE-TRANSITION (validation)
- N-TASK-COMPLETE-ALLOWED (decision)
- N-TASK-COMPLETE-ALREADY (refusal)
- N-TASK-COMPLETE-STORE (operation)
- N-TASK-COMPLETE-CONFLICT (refusal)
- N-TASK-COMPLETE-ERROR (failure)
- N-TASK-COMPLETE-DONE (result)
Edges:
- E-TASK-COMPLETE-START: N-TASK-COMPLETE-START -> N-TASK-COMPLETE-AUTH
- E-TASK-COMPLETE-DENIED: N-TASK-COMPLETE-AUTH -> N-TASK-COMPLETE-DENIED
- E-TASK-COMPLETE-ALLOWED: N-TASK-COMPLETE-AUTH -> N-TASK-COMPLETE-ID
- E-TASK-COMPLETE-1: N-TASK-COMPLETE-ID -> N-TASK-COMPLETE-VALID
- E-TASK-COMPLETE-2: N-TASK-COMPLETE-VALID -> N-TASK-COMPLETE-INVALID
- E-TASK-COMPLETE-3: N-TASK-COMPLETE-VALID -> N-TASK-COMPLETE-LOAD
- E-TASK-COMPLETE-4: N-TASK-COMPLETE-LOAD -> N-TASK-COMPLETE-MISSING
- E-TASK-COMPLETE-5: N-TASK-COMPLETE-LOAD -> N-TASK-COMPLETE-ERROR
- E-TASK-COMPLETE-6: N-TASK-COMPLETE-LOAD -> N-TASK-COMPLETE-TRANSITION
- E-TASK-COMPLETE-7: N-TASK-COMPLETE-TRANSITION -> N-TASK-COMPLETE-ALLOWED
- E-TASK-COMPLETE-8: N-TASK-COMPLETE-ALLOWED -> N-TASK-COMPLETE-ALREADY
- E-TASK-COMPLETE-9: N-TASK-COMPLETE-ALLOWED -> N-TASK-COMPLETE-STORE
- E-TASK-COMPLETE-10: N-TASK-COMPLETE-STORE -> N-TASK-COMPLETE-CONFLICT
- E-TASK-COMPLETE-11: N-TASK-COMPLETE-STORE -> N-TASK-COMPLETE-MISSING
- E-TASK-COMPLETE-12: N-TASK-COMPLETE-STORE -> N-TASK-COMPLETE-ERROR
- E-TASK-COMPLETE-13: N-TASK-COMPLETE-STORE -> N-TASK-COMPLETE-DONE

Complete task

GRAPH-TASK-COMPLETE



Edge key

E1 E-TASK-COMPLETE-START
E2 E-TASK-COMPLETE-DENIED — missing / unresolved / insufficient
E3 E-TASK-COMPLETE-ALLOWED — authorized
E4 E-TASK-COMPLETE-1
E5 E-TASK-COMPLETE-2 — fail
E6 E-TASK-COMPLETE-3 — pass
E7 E-TASK-COMPLETE-4 — NOT_FOUND
E8 E-TASK-COMPLETE-5 — unavailable / invalid state
E9 E-TASK-COMPLETE-6 — ready Task
E10 E-TASK-COMPLETE-7
E11 E-TASK-COMPLETE-8 — fail
E12 E-TASK-COMPLETE-9 — pass
E13 E-TASK-COMPLETE-10 — VERSION_CONFLICT
E14 E-TASK-COMPLETE-11 — NOT_FOUND
E15 E-TASK-COMPLETE-12 — STORE_UNAVAILABLE
E16 E-TASK-COMPLETE-13 — confirmed

Graph GRAPH-TASK-LIST

List tasks

Kind: flow

Nodes

Node N-TASK-LIST-START (start)

Receive request

Node N-TASK-LIST-AUTH (decision)

Trusted permission?

Requirement refs: RF-TASK-LIST-AUTH

Node N-TASK-LIST-DENIED (refusal)

Denied; no tool call

Requirement refs: RF-TASK-LIST-AUTH

Node N-TASK-LIST-READ (operation)

Read board snapshot

Call: operation OP-TASK-LIST

Requirement refs: RF-TASK-LIST-SNAPSHOT

Node N-TASK-LIST-FAIL (failure)

Read failure; no list

Requirement refs: RF-TASK-LIST-FAILURE

Node N-TASK-LIST-RETURN (result)

Return exact snapshot

Requirement refs: RF-TASK-LIST-SNAPSHOT

Edges

Edge E-TASK-LIST-START: N-TASK-LIST-START → N-TASK-LIST-AUTH

Requirement refs: none

Edge E-TASK-LIST-DENIED: N-TASK-LIST-AUTH → N-TASK-LIST-DENIED

Label

missing / unresolved / insufficient

Requirement refs: RF-TASK-LIST-AUTH

Edge E-TASK-LIST-ALLOWED: N-TASK-LIST-AUTH → N-TASK-LIST-READ

Label

authorized

Requirement refs: RF-TASK-LIST-AUTH

Edge E-TASK-LIST-1: N-TASK-LIST-READ → N-TASK-LIST-FAIL

Label

unavailable / invalid state

Requirement refs: RF-TASK-LIST-FAILURE

Edge E-TASK-LIST-2: N-TASK-LIST-READ → N-TASK-LIST-RETURN

Label

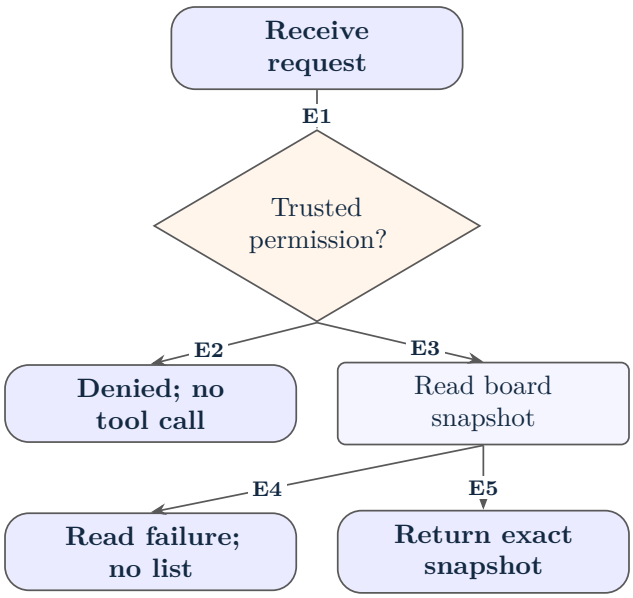
complete snapshot

Requirement refs: RF-TASK-LIST-SNAPSHOT
Topology of GRAPH-TASK-LIST:
Nodes:
- N-TASK-LIST-START (start)
- N-TASK-LIST-AUTH (decision)
- N-TASK-LIST-DENIED (refusal)
- N-TASK-LIST-READ (operation)
- N-TASK-LIST-FAIL (failure)
- N-TASK-LIST-RETURN (result)

Edges:
- E-TASK-LIST-START: N-TASK-LIST-START -> N-TASK-LIST-AUTH
- E-TASK-LIST-DENIED: N-TASK-LIST-AUTH -> N-TASK-LIST-DENIED
- E-TASK-LIST-ALLOWED: N-TASK-LIST-AUTH -> N-TASK-LIST-READ
- E-TASK-LIST-1: N-TASK-LIST-READ -> N-TASK-LIST-FAIL
- E-TASK-LIST-2: N-TASK-LIST-READ -> N-TASK-LIST-RETURN

List tasks

GRAPH-TASK-LIST



Edge key

E1 E-TASK-LIST-START
E2 E-TASK-LIST-DENIED — missing / unresolved / insufficient
E3 E-TASK-LIST-ALLOWED — authorized
E4 E-TASK-LIST-1 — unavailable / invalid state
E5 E-TASK-LIST-2 — complete snapshot

Appendix: assumptions

ASM-TASK-SINGLE-BOARD

Statement

Synthetic scope assumes one resolved board per invocation and stable task identifiers within that board.

Status: open

Affected: ENT-TASK, TOL-TASK-STORE, UC-TASK-CREATE, UC-TASK-COMPLETE, UC-TASK-LIST

ASM-TASK-IDENTITY

Statement

A future application has a trusted identity-and-permission authority that resolves principal, board, grants and validity. This capability is assumed, not implemented or empirically demonstrated here.

Status: open

Affected: UC-TASK-CREATE, UC-TASK-COMPLETE, UC-TASK-LIST

Appendix: sources

SRC-TASK-BRIEF

Title

Synthetic task-board teaching brief

Uri: needs.md

Revision: 1

Accessed: null

Note

Authored for this repository as a synthetic baseline. It is not an imported stakeholder request, research observation, or independently approved Needs artifact. The repository maintainer must name an accountable person before claiming a real Needs-to-L1 derivation.

SRC-TASK-OAL

Title

OAL Alpha-v0.2.0-Alkimim normative reference

Uri: https://github.com/CodeVaultCommunity/OAL-Open-Analysis-Layers/blob/ced9645dcdb1d1d367ac1b3c913256298255094e/versioning/Alpha-v0.2.0-Alkimim/README.md

Revision: ced9645dcdb1d1d367ac1b3c913256298255094e

Accessed: 2026-10-25

Note

Reference for layer boundaries, caller context, provenance and conformance distinctions. It does not independently supply the synthetic board's product requirements.

Appendix: history

HIST-TASK-BOARD-01

Version: 1

Date: 2026-10-25

Description

Initial reference example for the first Atlas Specifications edition, Alpha-v0.1.0. This synthetic example demonstrates the model and its representations; it does not assert stakeholder approval or execution of the described business application.

Derived summary: 5 sections; 1 actors; 1 entities; 1 tools; 3 use cases; 1 boundaries; 6 graphs; 0 resources.